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1. Executive Summary

The goal of this project is to use data taken from a Postgres database to conduct a thorough analysis of the performance of the company in question. The dataset has 13 interrelated tables that reflect numerous aspects of the company's activities, including customers, staff, products, orders, and suppliers. It is based on the well-known Northwind database, the fictitious company that manages a large and intricate database with numerous stakeholders, including suppliers, consumers, workers, and products, in several geographical regions. The organization must overcome a few challenges to effectively analyze this data in order to boost business performance. The main objective is to use SQL queries to provide answers to important business queries that shed light on the company's operations, sales figures, and customers by delivering data-driven, actionable insights to corporate stakeholders that could guide decision-making.

Without advanced querying techniques and appropriate data organization, it might be challenging to extract relevant insights from a large number of interconnected tables (Sarker, 2021). To ensure smooth operations and continued growth, it is essential to have the answers to important business issues, such as identifying top-performing items, comprehending client demographics, and tracking order fulfillment (Lemon & Verhoef, 2016).

After defining the foreign key and primary key relationships using SQL queries, the analysis generated informative results for a range of departmental business purposes. In order to help the Product Team evaluate offerings, a list of products with prices between \$20 and \$50 was identified. This list classified products by price and showed those that were not discontinued. Brazil and France were identified as major regions based on logistics performance in 1998, which highlighted countries with more than 10 orders and shipping delays longer than five days. HR insights aided in tracking career growth included the employee's age at employment and their reporting structure. According to global logistics performance from 1997 to 1998, the best month in terms of freight volume was April 1998. In order to help with pricing decisions, products with notable price changes outside of the 20%–30% range were marked.

Key insights for sales volume and discounted totals were revealed by category performance throughout price ranges, and regional supplier stock data assisted in determining the amount of inventory required by region. Competitive pricing was supported by the comparison of product prices to category averages, which showed which products were above or below their category benchmarks. Employee sales KPIs showed Margaret Peacock and other top achievers with specific performance measures. Ultimately, individual contributions within each area were highlighted by employee sales across categories, which supported performance reviews and the optimization of sales strategies.



2. Introduction

This main goal of the project is to give data-driven analysis to improve decision-making in the company's pricing, logistics, and sales departments, among other departments. The project's goal was to produce useful insights that could be used to manage supplier relationships, measure employee performance, and optimize product pricing strategies. The main objective was to provide the business with the knowledge and resources needed to boost revenue, increase operational effectiveness, and increase worker productivity.

- **Key Stakeholders and Their Requirements**

- **Pricing Team:** In order to evaluate product price across several categories, spot anomalies, and make sure the company's pricing strategy satisfies consumer demand, the pricing team needed thorough analysis. They specifically intended to examine products with significant price variations and compare the costs of particular products to the averages and medians of the category.
 - Addressing requirements: By comparing the unit price of each product to the average and median pricing in the category, the analysis was able to pinpoint outliers and provide a more comprehensive picture of product positioning. This assisted the group in fine-tuning pricing tactics to correspond with market dynamics and consumer segmentation.
- **Logistics Team:** The Logistics Team required information on stock levels, order fulfillment efficiency, and regional supplier performance. In order to effectively manage supply chain activities, they sought to pinpoint places where shipment delays happened and evaluate the present level of inventory across supplier regions.
 - Addressing requirements: Through an examination of supplier regions, stock levels, and freight efficiency, the research identified potential operational risks associated with stock shortages or shipment delays. As a result, the logistics team was able to increase supply chain efficiency and manage inventory proactively.
- **Sales Team:** The sales team used average order values, discount usage, and total sales as key performance indicators (KPIs) to assess worker performance. To pinpoint high-achieving staff members and places where sales performance could be maximized, they needed precise measurements.

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- Addressing requirements: A thorough breakdown of KPIs, such as total sales (with and without discounts), average order value, and discount usage, was provided by the employee performance analysis. The sales team was able to benchmark worker performance, reward top performers, and pinpoint areas in need of development thanks to these indicators.

To sum up, the project has been successful in providing data-driven insights that have been customized to the needs of the sales, pricing, and logistics teams. These insights enabled the teams to make decisions that would enhance performance and spur growth.



3. Data Understanding

- **Dataset Overview and Sources:**

The dataset is derived from a fictitious company called Northwind Database. It replicates the operations of a company that deals with goods, clients, vendors, staff, and orders. With 13 tables that cover several aspects of the company's activities, it offers an extensive amount of data for investigating and evaluating the effectiveness of the company. Data on product categories, customers, employees, orders, suppliers, territories, and other information are included in the tables.

A PostgreSQL database stores the data in a relational structure with tables linked by the primary and foreign key relationships. This framework allows complicated queries to provide answers to a range of business concerns and facilitates the efficient retrieval of information.

- **Data Sources and Collection Methods:**

The dataset collects data on sales, customer interactions, inventory management, and personnel information, all of which are modeled following real-world business environments.

Limitations:

- Temporal Restrictions: All transactions in the dataset are historical, and real-time data is not included. This can make it harder to see how well businesses are doing currently.
- Data Completeness: For some variables, such as ship dates or client demographics, there are missing or insufficient data.
- Fictional Nature: Some assumptions regarding the company's operations may not precisely match real-world circumstances as the dataset is fictional.

- **Variables/features in the dataset and their significance:**

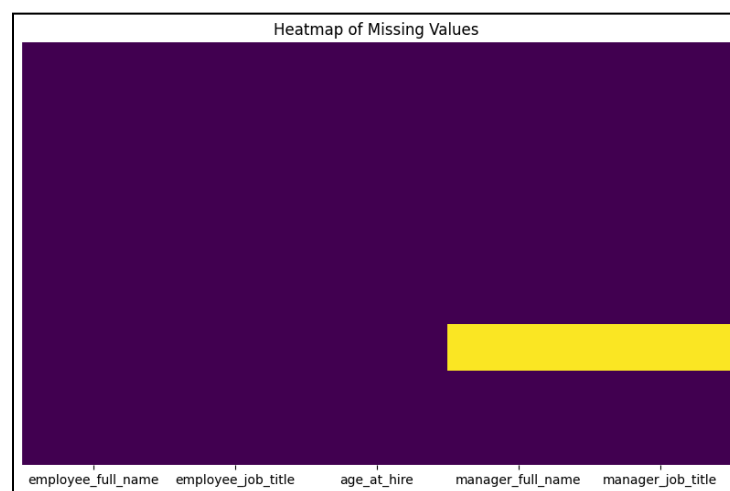
- Orders and Order Details: Customer interactions, products ordered, quantities, discounts, and shipment information are all recorded in these tables.
- Employees and Employee Territories: Employee roles, regional assignments, and reporting structures are all included in these tables that outline the company's human resources. Employee territories are also included.

- Products, Categories, and Suppliers: The information in these tables pertains to the products that the business sells, how they are classified, and whose suppliers are in charge of supplying them.
- Customers and Demographics: These tables provide information about the company's names, contacts, and geographic details of the customers, as well as demographic information where applicable.
- **Exploratory Data Analysis (EDA):**

Preliminary data exploration has been performed to gain a better understanding of the dataset, and the results provided important insights into its distribution and structure.

- Fifteen tables are included in the dataset. There is more data in important tables like Orders (830 rows), OrderDetails (2155 rows), and Products (77 rows) that needs to be examined more closely.
- The dataset is made up of a variety of data types, such as dates (OrderDate, HireDate), numbers (EmployeeID, OrderID), strings (CustomerName, CategoryName), and monetary values (Freight, UnitPrice).
- Primary and foreign keys have been defined to provide relationships between tables, enabling more intricate queries.
- Upon preliminary examination of the data, certain missing values were found, mostly in the shipment (ShipDate) and manager's details from the Employees table (Figure 1), which could potentially affect the delivery performance analysis.

Figure 1. Heatmap of Missing Values in Manager's Details



- **Key Takeaways from EDA:**

- Sales Performance: The basis for examining the company's sales patterns and product performance is provided by the Orders and OrderDetails tables. A thorough examination of consumer preferences and product profitability will be possible by combining them with customers and products.
- Geographic Distribution: By examining sales and operations by region and determining which regions contribute more to sales, one can examine the Customers, Employees, Territories, and Regions tables.
- Employee Performance: One can assess the sales performance of their employees and look into how employee territories affect sales by connecting the Orders and Employees tables.



4. Business Question 1

a. Description

A list of items with unit values between \$20 and \$50 was sought by the Product Team as part of their annual analysis of the company's pricing policy. Products that fall inside this price range and are still being sold are of special interest to them. To list the most costly products first, the results should be sorted by unit price in descending order. The team will use this analysis to examine the mid-range pricing category and determine whether the pricing strategy needs to be modified.

b. Results

The products that fit the criteria (unit price between \$20 and \$50 and not discontinued) are listed below based on the first business query. The products are arranged from highest to lowest unit price:

Table 1. List of products (unit price between \$20 and \$50 and not discontinued)

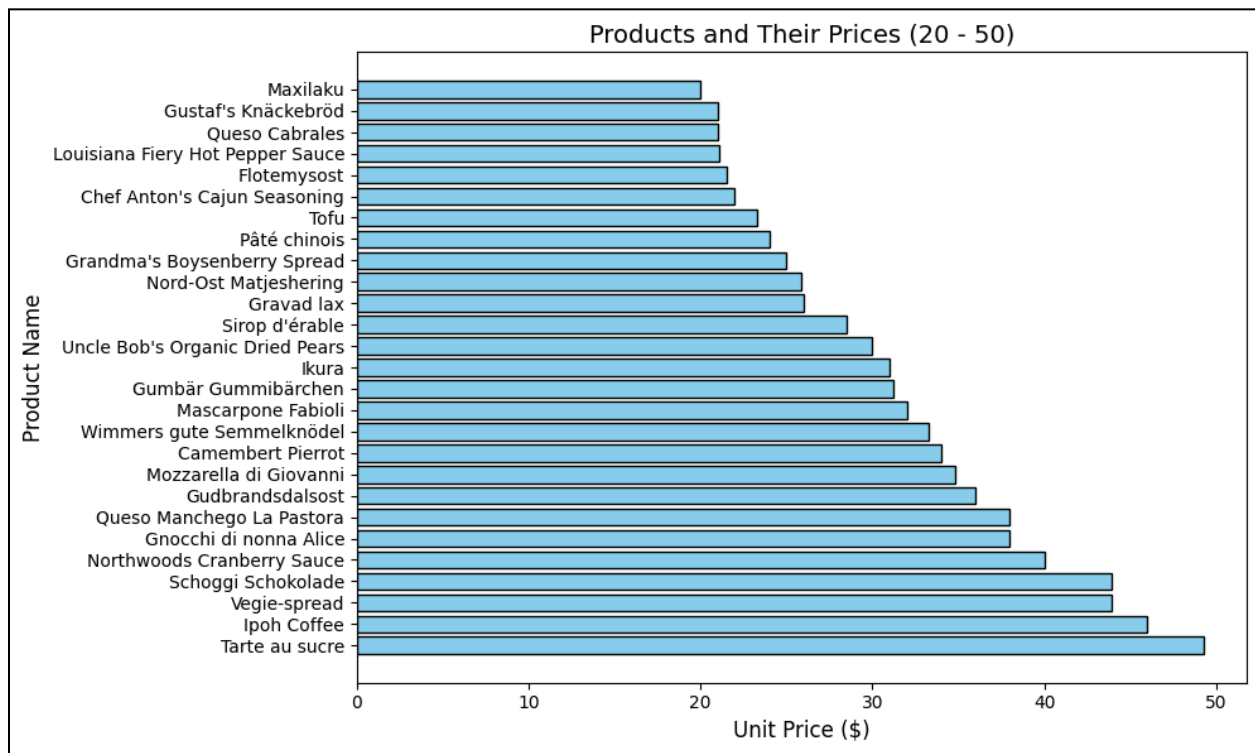
Product Name	Unit Price (\$)
Tarte au sucre	49.3
Ipoh Coffee	46.0
Vegie-spread	43.9
Schoggi Schokolade	43.9
Northwoods Cranberry Sauce	40.0
Gnocchi di nonna Alice	38.0
Queso Manchego La Pastora	38.0
Gudbrandsdalsost	36.0
Mozzarella di Giovanni	34.8
Camembert Pierrot	34.0
Wimmers gute Semmelknödel	33.25
Mascarpone Fabioli	32.0
Gumbär Gummibärchen	31.23
Ikura	31

Uncle Bob's Organic Dried Pears	30
Sirop d'érable	28.5
Gravad lax	26
Nord-Ost Matjeshering	25.89
Grandma's Boysenberry Spread	25
Pâté chinois	24
Tofu	23.25
Chef Anton's Cajun Seasoning	22
Flotemysost	21.5
Louisiana Fiery Hot Pepper Sauce	21.05
Queso Cabrales	21
Gustaf's Knäckebröd	21
Maxilaku	20

c. Key insights and Findings

- **Pricing Distribution:** There appears to be a wide range of products in this pricing category, as evidenced by the unit costs in this range, which range from \$21.05 to \$49.30. "Maxilaku" is one of the less expensive products, while "Tarte au sucre" and "Ipoh Coffee" are at the highest end of the pricing spectrum (Figure 2).
- **Diversity in Product Offering:** There are many different products available in this price category, ranging from baked goods (such as "Tarte au sucre") to beverages (such as "Ipoh Coffee") and gourmet foods (such as "Mishi Kobe Niku"). This suggests that a wide number of product categories are included in the \$20 to \$50 price range.
- **Strategic Importance:** Considering their positioning between lower- and higher-priced products, these midpriced commodities have the potential to be high-margin products. The team may concentrate on this spectrum in order to increase profitability and keep prices competitive.

Figure 2. Continued Products and Their Prices (\$20- \$50)



- **Future Recommendations:** Based on this study, the Product Team can decide if any price changes are necessary to move any products into the premium category (above \$50) or to think about implementing promotional pricing plans for goods that are getting close to the \$20 price threshold.

In order to keep the business competitive in the market, the product team can use the aforementioned results as a guide for assessing their pricing strategy for this market segment.



5. Business Question 2

a. Description

In order to identify opportunities for improvement—especially in countries where their logistics performance fell short of expectations—the Logistics Team is looking back at their 1998 performance. They asked for a list of countries with at least ten distinct orders placed in a given year with an average shipping delay of five days or more (determined by dividing the order date by the shipping date). The nation names should be used to sort the results alphabetically.

b. Results

Instructions: Explain the results of your analysis in a non-technical manner. Include the table with data that directly answers the business question.

Table 2. Average Days Between Order and Shipping for Total Unique Orders by Country (1998)

Country	Average Days Between Order and Shipping	Total Unique Orders
Austria	5.89	11
Brazil	8.12	28
France	9.43	23
Germany	5.38	34
Spain	7.83	12
Sweden	13.29	14
UK	6.25	16
USA	7.89	39
Venezuela	8.73	18

c. Key insights and Findings

- **Countries with High Average Shipping Delays:**
 - In 1998, Sweden had the greatest average shipping delay of 13.29 days, suggesting possible logistical difficulties in this area.

- With an average delay of 9.43 days, France trailed closely behind, indicating that it too had considerable shipping delays in comparison to other nations.
- **Countries with Better Performance:**
 - Although the average shipment delays for Germany and Austria were less than the 5-day barrier that the Logistics Team wants to look into, they were still 5.38 days and 5.89 days, respectively.
 - Spain experienced a noteworthy 7.83-day delay as well.
- **Number of Orders:**
 - With 34 unique orders in 1998, Germany was the region with the highest number, making it a top target for increasing delivery efficiency.
 - Brazil placed 28 distinct purchases, highlighting the region's high order volume and longer shipment periods as reasons for shipping delays that need to be addressed.

Figure 3. Average days between orders and shipping with the total unique orders for each country in 1998



These findings show that, although shipping delays are a problem in every country on the list, areas like Brazil and France stand out because of their large order volumes and higher average delays. The logistics team may improve performance, cut down on delays, and refine their shipping strategy in these failing regions by using the insights from this retrospective research. Figure 3 compares the average days between orders and shipping with the total unique orders for each country side by side for the year 1998.

6. Business Question 3

a. Description

The HR Team asked for information regarding each employee's reporting structure as well as an analysis of their age when they joined the company to examine the workforce and reporting hierarchy of the organization. They are specifically looking for a list of all employees that has the following details:

- Full name (first and last names combined)
- Job title
- Age at the time of hire
- Full name of the employee's manager (if applicable)
- Job title of the manager

The employee's age at the time of hire and their full name should be arranged by the results in ascending order. With this analysis, the HR team can evaluate leadership skills and management duties to better understand the workforce demographics and reporting lines of the company.

b. Results

The results are included in the table below, which also contains information about each employee's boss and their full name, job title, and age at hire.

Table 3. Employee's full name, job title, age at hire, and details about their manager

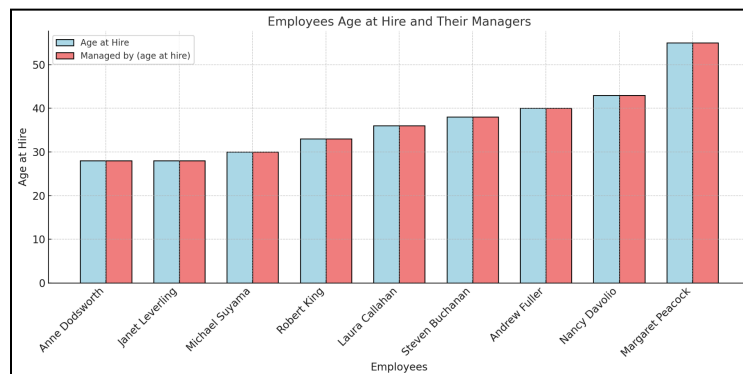
Employee Full Name	Job Title	Age at Hire	Manager Full Name	Manager Job Title
Anne Dodsworth	Sales Representative	28	Steven Buchanan	Sales Manager
Janet Leverling	Sales Representative	28	Andrew Fuller	Vice President, Sales
Michael Suyama	Sales Representative	30	Steven Buchanan	Sales Manager
Robert King	Sales Representative	33	Steven Buchanan	Sales Manager
Laura Callahan	Inside Sales Coordinator	36	Andrew Fuller	Vice President, Sales

Steven Buchanan	Sales Manager	38	Andrew Fuller	Vice President, Sales
Andrew Fuller	Vice President, Sales	40	NULL	NULL
Nancy Davolio	Sales Representative	43	Andrew Fuller	Vice President, Sales
Margaret Peacock	Sales Representative	55	Andrew Fuller	Vice President, Sales

c. Key insights and Findings

- **Age at Hire:** The results show that the employees in this sample were hired between the ages of 28 and 55. This implies that the organization employs people at various phases of their careers, with both younger and more experienced professionals entering the team.
- **Organizational Structure:** Many employees are subordinate to senior managers like Vice President of Sales Andrew Fuller and Sales Manager Steven Buchanan. This makes it easy to comprehend the sales department's hierarchical reporting structure within the organization.
- **Distribution of Employees:** The data shows a trend where a manager oversees several employees, like Steven Buchanan, who oversees multiple sales representatives. This pattern can assist the HR staff in determining how managers divide up the workload.

Figure 4. Age at hire for employees alongside their managers



The grouped bar chart in Figure 4 compares between employees' ages when they joined the company and the managers they report to. It visually emphasizes the hierarchical relationships while also indicating the age distribution at the time of hiring.



7. Business Question 4

a. Description

A retrospective examination of the logistics team's worldwide performance during 1997 and 1998 is being carried out. The months that did well in terms of order volume and freight totals are of special importance to them. The group asked for a list of the following monthly performance metrics:

- The year and month (formatted as a single date field)
- The total number of orders for each month
- The total freight amount for each month (rounded to the nearest whole number)

The result should be arranged in descending order of total orders, and the study was limited to the months in which the total number of orders exceeded 35. This will assist the team in identifying months with strong performance and assessing freight expenses during those times.

b. Results

Instructions: Explain the results of your analysis in a non-technical manner. Include the table with data that directly answers the business question.

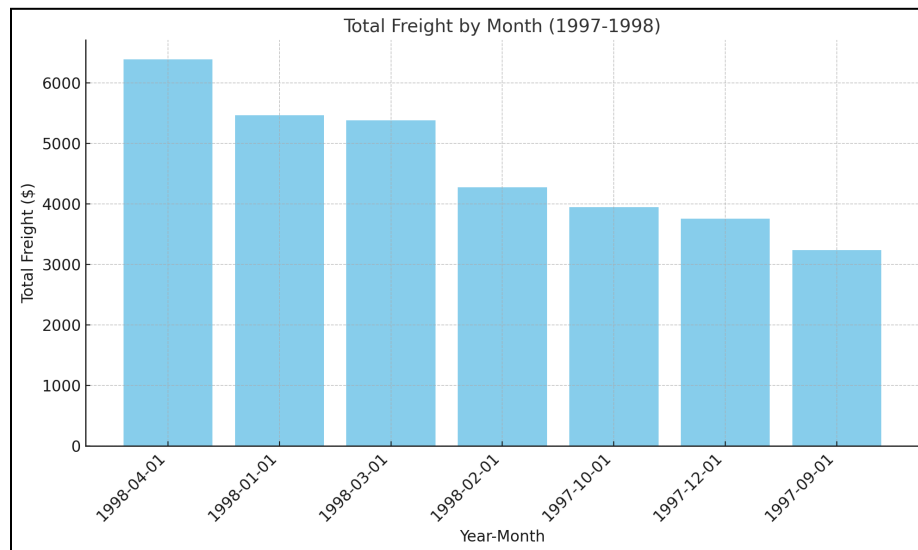
Table 4. Top-performing months based on total freight and total orders between 1997 and 1998

Year-Month	Total Orders	Total Freight
1998-04-01	74	6,394
1998-01-01	55	5,463
1998-03-01	73	5,379
1998-02-01	54	4,273
1997-10-01	38	3,946
1997-12-01	48	3,758
1997-09-01	37	3,237

c. Key insights and Findings

- Top-Performing Months:
 - April 1998 was the most significant month in terms of both order volume and freight, with the greatest overall freight cost of \$6,394 and 74 orders.
 - January 1998 was again a successful month, with 55 orders and \$5,463 in goods.
- Consistency in 1998:
 - The first four months of 1998 showed strong performance, with notable order volumes and high total freight values every month. This implies that in the first few months of 1998, the company faced significant demand for its logistics services.
- 1997 performance:
 - With 38 orders and \$3,946 in goods, October 1997 was one of the greatest months of the year, even though the total amount of goods was less than in the first few months of 1998.

Figure 5. Total Freight by Month (1997-1998)



According to this analysis, the logistics team performed at its best early in 1998, especially in April (Figure 5). They can optimize resource allocation and future logistical planning by comprehending this pattern, which is based on historical performance.

8. Business Question 5

a. Description

The Pricing Team is looking for products that have had price increases, but not by a proportion that falls between 20% and 30%. They demand a list of products with price increases that are either more (greater than 30%) or lower (less than 20%) than anticipated. The group requested the following information for every product:

- Product name
- Current unit price (formatted to two decimal places)
- Initial unit price (formatted to two decimal places)
- Percentage increase $\rightarrow ((\text{CurrentUnitPrice} - \text{InitialUnitPrice}) / \text{InitialUnitPrice} * 100)$
- Current unit price (formatted to two decimal places)
- Initial unit price (formatted to two decimal places)

To make it easier to identify the goods with the biggest and smallest deviations, the results are arranged in increasing order of percentage increase.

b. Results

The products whose price rises fell outside of the predicted range are shown in the following table. For instance, the unit price of "Queso Cabrales" increased by 50% and "Singaporean Hokkien Fried Mee" increased by 43%, both of which are more than the 30% criterion.

Table 5. Products that did not fall within the 20-30% price increase range

Product Name	Current Unit Price	Initial Unit Price	Percentage Increase (%)
Singaporean Hokkien Fried Mee	14.00	9.80	43
Queso Cabrales	21.00	14.00	50

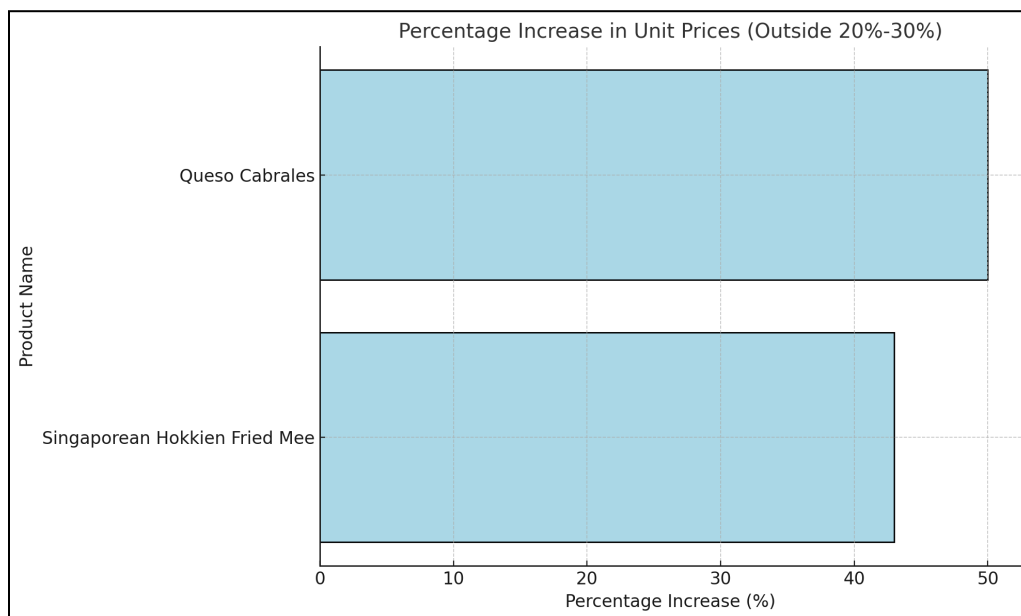
c. Key insights and Findings

- **Notable Price Increases:** The two products in the result saw significant price rises. The price increases for "Queso Cabrales" and "Singaporean Hokkien Fried Mee" were 50%

and 43%, respectively, much more than the highest limit of 30%. This suggests that additional research may be necessary for some products in order to ascertain whether the price rises are consistent with the market or whether they are abnormal.

- **No Products with Lower Price Increases:** Only notable price increases are highlighted in the sample; no product has a price rise of less than 20%. The Pricing Team will be able to concentrate on products whose price alterations may merit more investigation, nevertheless, thanks to our analysis.

Figure 6. Percentage Increase in Unit Prices (Outside 20%-30%)



9. Business Question 6

a. Description

The Pricing Team seeks to determine how well product categories are performing in relation to their price points. The group asked for a summary of the categories, price ranges that correspond to them, total income (after deducting discounts), and the quantity of orders that fell into each category. The following categories apply to the price ranges:

- "1. Below \$20"
- "2. \$20-\$50"
- "3. Over \$50"

The team gains a better understanding of how various price points within each category affect overall sales volume and performance with the aid of this data.

b. Results

The results are shown in Table 6, which includes the categories, pricing ranges, total amount (after discount), and number of orders for each category within the designated price ranges:

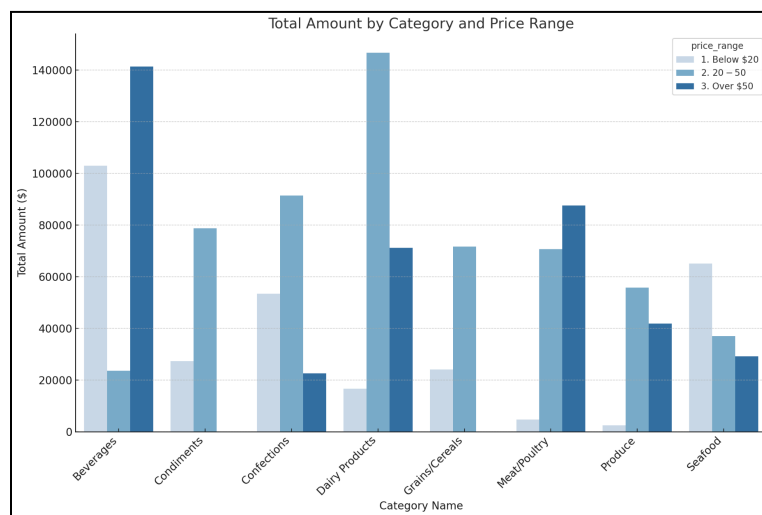
Table 6. Price ranges within categories in terms of total revenue and order volume

category_name	price_range	total_amount	volume_of_orders
Beverages	1. Below \$20	102945	317
Beverages	2. \$20 - \$50	23527	28
Beverages	3. Over \$50	141397	24
Condiments	1. Below \$20	27299	85
Condiments	2. \$20 - \$50	78748	121
Confections	1. Below \$20	53365	197
Confections	2. \$20 - \$50	91429	106
Confections	3. Over \$50	22563	16
Dairy Products	1. Below \$20	16569	81
Dairy Products	2. \$20 - \$50	146783	204
Dairy Products	3. Over \$50	71156	54
Grains/Cereals	1. Below \$20	24071	99
Grains/Cereals	2. \$20 - \$50	71673	91
Meat/Poultry	1. Below \$20	4728	36
Meat/Poultry	2. \$20 - \$50	70699	96
Meat/Poultry	3. Over \$50	87595	36
Produce	1. Below \$20	2432	13
Produce	2. \$20 - \$50	55732	81
Produce	3. Over \$50	41820	39
Seafood	1. Below \$20	65110	217
Seafood	2. \$20 - \$50	36980	70
Seafood	3. Over \$50	29172	27

c. Key insights and Findings

- **High Quality Performance at a Reasonable Cost:** The amount of sales and order volume for ‘Beverages’ and ‘Condiments’ originate from items with prices under \$20. For instance, 317 orders for beverages in the “Below \$20” area brought in \$102,945 in revenue.
- **Considerable Impact from the Higher Price Range:** Products in the “Over \$50” price range contribute significantly to total revenue, even if they are not as often ordered. For example, “Beverages,” costing more than \$50, generated \$141,397 in revenue from just 24 orders, demonstrating the viability of expensive products even in the face of lower sales volume.
- **Mid-Range Products:** \$20 to \$50 products have varying levels of performance in different categories. For example, in this price range, “Condiments” had 121 orders and made \$78,748; in contrast, “Beverages” had only 28 orders in the same range and made \$23,527.

Figure 7. Total Amount by Category and Price Range for the categories and respective price ranges



It is simple to compare the revenue distribution across price segments because each bar shows the total amount generated for a certain price range within a category (Figure 7). By concentrating on the most lucrative price ranges within each category, the Pricing Team can use these results to inform future pricing plans and marketing initiatives.



10. Business Question 7

a. Description

An examination of the current stock levels of regional suppliers for each product category was requested by the logistics team. They wish to evaluate the total number of units in stock, total number of units on order, and total amount of reorders in three different regions:

- America
- Europe
- Asia-Pacific

The logistics team will be able to comprehend stock levels and possible supply chain threats for each region and category with the aid of this information. To make analysis easier, the results are sorted in ascending order by supplier location, category name, and ordering level.

b. Results

The following table represents the current stock status for categories across different supplier regions:

Table 7. Current Stock Levels of Regional Suppliers for Each Product Category

supplier_region	category_name	total_units_in_stock	total_units_on_order	total_reorder_level
America	Beverages	20	0	0
Asia-Pacific	Beverages	215	20	95
Europe	Beverages	324	40	100
America	Condiments	113	0	25
Asia-Pacific	Condiments	349	100	70
Europe	Condiments	45	70	40
America	Confections	17	0	0
Asia-Pacific	Confections	29	0	10
Europe	Confections	340	180	155
Europe	Dairy Products	393	140	110
Asia-Pacific	Grains/Cereals	64	0	25
Europe	Grains/Cereals	244	90	130
America	Meat/Poultry	136	0	30
Asia-Pacific	Meat/Poultry	29	0	0
Europe	Meat/Poultry	0	0	0
Asia-Pacific	Produce	74	20	25
Europe	Produce	26	0	0
Asia-Pacific	Seafood	305	0	45
Europe	Seafood	396	120	100

c. Key insights and Findings

- **Beverages Category:**

- With 324 units in stock and 40 units on order, Europe has the greatest beverage stock, while America has the smallest stock with only 20 units and no orders.
- Asia-Pacific is positioned in the center, having 95 units on reorder and 215 units in stock. 20 units are also on order for this region.

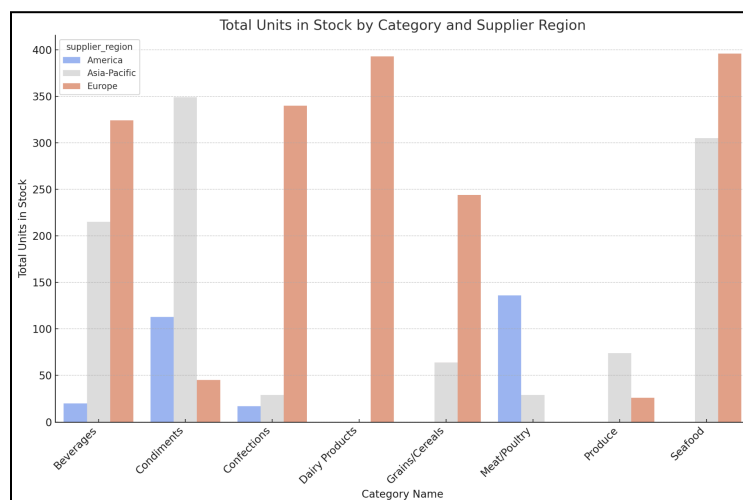
- **Category of Condiments:**

- With 349 units in stock and 100 units on order, Asia-Pacific has more condiments than any other region; in contrast, America has 113 units and no open orders.
- In terms of condiment reorder levels, America has the lowest level—25 units—while Asia-Pacific has the highest—70 units.

- **Analysis of Reorder Level:**

- The reorder levels differ greatly by area, with Europe and Asia-Pacific typically having greater reorder levels. This could be a symptom of more cautious or aggressive supply chain management in these regions as opposed to America.

Figure 8. Total Units In Stock By Category And Supplier Region



The logistics team may rapidly evaluate stock distribution by using the bar chart in Figure 8, which illustrates how stock levels vary by region for each product type.



11. Business Question 8

a. Description

In order to determine how the unit prices of all currently available products compare to the average and median unit costs within their respective categories, the Pricing Team requested an analysis of all currently provided products. Every product is broken down in great detail in the analysis, along with its name, unit price, category average and median pricing, and ranking in relation to these criteria. Only products that are still in production are considered for analysis, and the findings are arranged alphabetically by product name and category. This business question seeks to evaluate:

- Each product's **category name** and **product name**.
- **Unit price** of the product.
- The **average** and **median unit price** for the product's category.
- The product's **position relative to the category's average unit price** (Below Average, Equal Average, Over Average).
- The product's **position relative to the category's median unit price** (Below Median, Equal Median, Over Median).

b. Results

The table below presents the results of all currently offered products and their unit prices compared to the average and median unit prices within their respective categories. In the following table:

- With a unit price of \$18, "Chartreuse verte" is exactly in line with the median but less expensive than the category average of \$46.03.
- With a price of \$263.50, which is substantially more than the 'Beverages' category's average and median, "Côte de Blaye" stands out.
- With a unit price of \$14, "Laughing Lumberjack Lager" is less expensive than the median and average.

Table 8. Currently Offered Products and their Unit Prices Compared to the Average and Median Unit Prices within respective Categories

category_name	product_name	unit_price	avg_unit_price	median_unit_price	price_vs_avg	price_vs_median
Beverages	Chartreuse verte	18	46.03	18	Below Average	Equal Median
Beverages	Côte de Blaye	263.5	46.03	18	Over Average	Over Median
Beverages	Ipoh Coffee	46	46.03	18	Below Average	Over Median
Beverages	Lakkalikööri	18	46.03	18	Below Average	Equal Median
Beverages	Laughing Lumberjack Lager	14	46.03	18	Below Average	Below Median
Beverages	Outback Lager	15	46.03	18	Below Average	Below Median
Beverages	Rhönbräu Klosterbier	7.75	46.03	18	Below Average	Below Median
Beverages	Sasquatch Ale	14	46.03	18	Below Average	Below Median
Beverages	Steeleye Stout	18	46.03	18	Below Average	Equal Median
Condiments	Aniseed Syrup	10	22.99	21.05	Below Average	Below Median
Condiments	Chef Anton's Cajun Seasoning	22	22.99	21.05	Below Average	Over Median
Condiments	Genen Shouyu	13	22.99	21.05	Below Average	Below Median
Condiments	Grandma's Boysenberry Spread	25	22.99	21.05	Over Average	Over Median
Condiments	Gula Malacca	19.45	22.99	21.05	Below Average	Below Median
Condiments	Louisiana Fiery Hot Pepper Sauce	21.05	22.99	21.05	Below Average	Below Median
Condiments	Louisiana Hot Spiced Okra	17	22.99	21.05	Below Average	Below Median
Condiments	Northwoods Cranberry Sauce	40	22.99	21.05	Over Average	Over Median
Condiments	Original Frankfurter grüne Soße	13	22.99	21.05	Below Average	Below Median
Condiments	Sirup d'érable	28.5	22.99	21.05	Over Average	Over Median
Condiments	Veggie-spread	43.9	22.99	21.05	Over Average	Over Median
Confections	Chocolade	12.75	25.16	16.25	Below Average	Below Median
Confections	Gumbär Gummibärchen	31.23	25.16	16.25	Over Average	Over Median
Confections	Maxilaku	20	25.16	16.25	Below Average	Over Median
Confections	NuNuCa Nuß-Nougat-Creme	14	25.16	16.25	Below Average	Below Median
Confections	Pavlova	17.45	25.16	16.25	Below Average	Over Median
Confections	Schoggi Schokolade	43.9	25.16	16.25	Over Average	Over Median
Confections	Scottish Longbreads	12.5	25.16	16.25	Below Average	Below Median
Confections	Sir Rodney's Marmalade	81	25.16	16.25	Over Average	Over Median
Confections	Sir Rodney's Scones	10	25.16	16.25	Below Average	Below Median
Confections	Tarte au sucre	49.3	25.16	16.25	Over Average	Over Median
Confections	Teatime Chocolate Biscuits	9.2	25.16	16.25	Below Average	Below Median
Confections	Valkoinen suklaa	16.25	25.16	16.25	Below Average	Equal Median
Confections	Zaanse koeken	9.5	25.16	16.25	Below Average	Below Median
Dairy Products	Camembert Pierrot	34	28.73	33	Over Average	Over Median
Dairy Products	Flotemysost	21.5	28.73	33	Below Average	Below Median
Dairy Products	Gelato	2.5	28.73	33	Below Average	Below Median
Dairy Products	Gorgonzola Telino	12.5	28.73	33	Below Average	Below Median
Dairy Products	Goudbrandsdalsost	36	28.73	33	Over Average	Over Median
Dairy Products	Mascarpone Fabioli	32	28.73	33	Over Average	Below Median
Dairy Products	Mozzarella di Giovanni	34.8	28.73	33	Over Average	Over Median
Dairy Products	Queso Cabrales	21	28.73	33	Below Average	Below Median
Dairy Products	Queso Manchego La Pastora	38	28.73	33	Over Average	Over Median
Dairy Products	Raclette Courdavault	55	28.73	33	Over Average	Over Median
Grains/Cereals	Filo Mix	7	21.29	20.25	Below Average	Below Median
Grains/Cereals	Gnocchi di nonna Alice	38	21.29	20.25	Over Average	Over Median
Grains/Cereals	Gustaf's Knäckebröd	21	21.29	20.25	Below Average	Over Median
Grains/Cereals	Ravioli Angelo	19.5	21.29	20.25	Below Average	Below Median
Grains/Cereals	Turnbröd	9	21.29	20.25	Below Average	Below Median
Grains/Cereals	Wimmers gute Semmelknödel	33.25	21.29	20.25	Over Average	Over Median
Meat/Poultry	Pâté chinois	24	15.72	15.72	Over Average	Over Median
Meat/Poultry	Tourtière	7.45	15.72	15.72	Below Average	Below Median
Produce	Longlife Tofu	10	29.06	26.63	Below Average	Below Median
Produce	Marjimpup Dried Apples	53	29.06	26.63	Over Average	Over Median
Produce	Tofu	23.25	29.06	26.63	Below Average	Below Median
Produce	Uncle Bob's Organic Dried Pears	30	29.06	26.63	Over Average	Over Median
Seafood	Boston Crab Meat	18.4	20.68	16.7	Below Average	Over Median
Seafood	Camaron von Tigers	62.5	20.68	16.7	Over Average	Over Median
Seafood	Escargots de Bourgogne	13.25	20.68	16.7	Below Average	Below Median
Seafood	Gravad lax	26	20.68	16.7	Over Average	Over Median
Seafood	Ikura	31	20.68	16.7	Over Average	Over Median
Seafood	Inlagd Sill	19	20.68	16.7	Below Average	Over Median
Seafood	Jack's New England Clam Chowder	9.65	20.68	16.7	Below Average	Below Median
Seafood	Konbu	6	20.68	16.7	Below Average	Below Median
Seafood	Nord-Ost Matjeshering	25.89	20.68	16.7	Over Average	Over Median
Seafood	Röd Kaviar	15	20.68	16.7	Below Average	Below Median
Seafood	Rogede sild	9.5	20.68	16.7	Below Average	Below Median
Seafood	Spegesild	12	20.68	16.7	Below Average	Below Median

c. Key insights and Findings

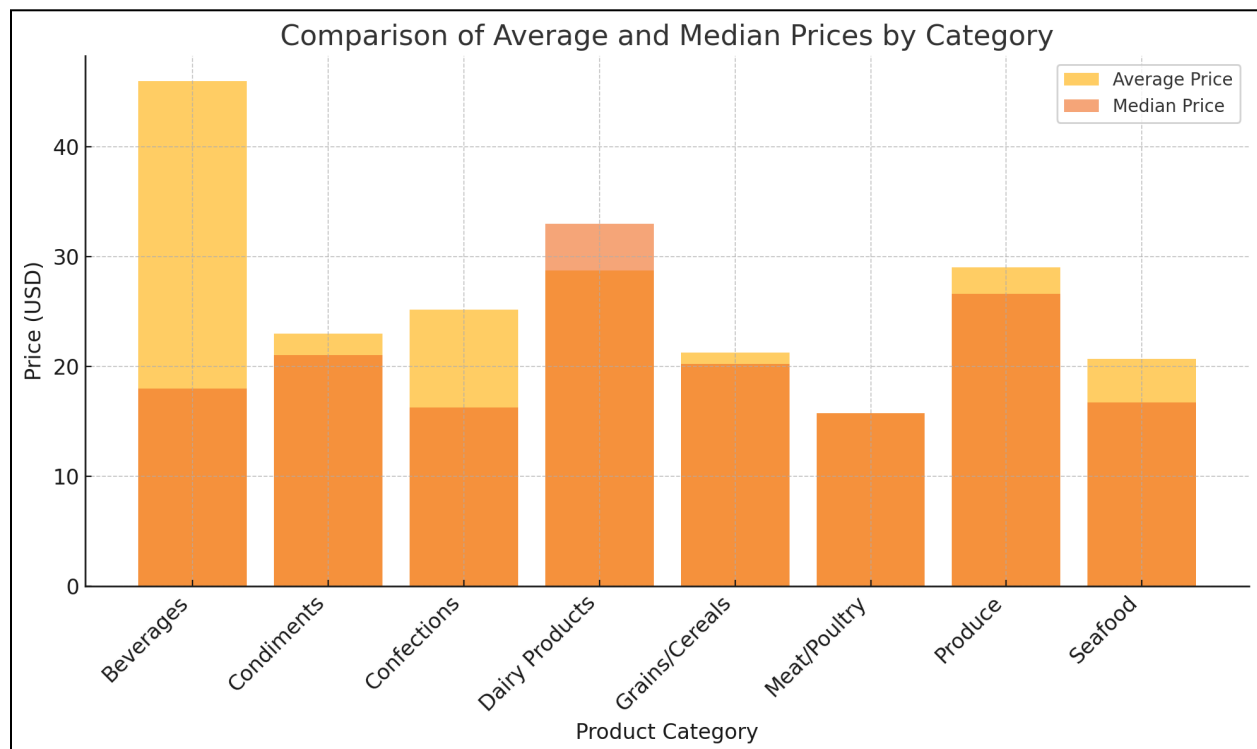
- **Pricing Dispersion:** Beverages is one of the categories with a large price range and notable outliers (e.g., Côte de Blaye at \$263.50). This implies different product demand or quality within the same category.
- **Positioning in Relation to Median:** A number of items are less expensive than the median in the category. It is crucial that the pricing team assesses whether or not their target

consumer segment is better served by less expensive products or if changes need to be made.

- **Overpriced Outliers:** Products with prices far higher than the median and average, such as Côte de Blaye, may require additional analysis to see if their premium status is due to strong brand positioning or demand.
- **Similar or Near the Median:** At the median price point, products like Lakkalikööri and Chartreuse verte are positioned neutrally within their category and might appeal to a wider range of buyers.

By utilizing pricing amounts to increase competitiveness or modifying outliers, the Pricing Team can utilize this data to fine-tune their product pricing strategy.

Figure 9. Comparison of Average and Median Prices by Category



The bar chart in Figure 9 contrasts the median and average prices of various product categories. This graph makes it easy to see the median price and average price within each category comparison.



12. Business Question 9

a. Description

Creating key performance indicators (KPIs) to assess staff performance is the goal that the sales team is interested in achieving. In order to aid in this endeavor, they asked for a list of workers along with a number of sales-related performance indicators, such as average sales amounts, percentages of discounts, and total sales amounts (both with and without discounts). The sales team will be able to determine high performers, evaluate the success of discount tactics, and comprehend the total sales contribution of each individual with the aid of this list.

- Employee full name
- Job title
- Total sales amount excluding discounts
- Total number of unique orders
- Total number of orders
- Average product amount excluding discounts
- Average order amount excluding discounts
- Total discount amount
- Total sales amount, including discounts
- Total discount percentage

The results are ordered by total sales amount, including discounts, in descending order.

b. Results

The sales team can identify top performers and gauge how well they manage orders and sales by looking at Table 9, which displays key performance data for staff members.

- With \$250,187.45 in sales before discounts and about \$232,890.85 after discounts, Margaret Peacock leads in both total sales with discounts and total sales with discounts combined. She handled 156 distinct orders.

- Even though he is the Vice President of Sales, Andrew Fuller performs well, handling more sales with fewer unique orders and a higher average order amount of \$1,851.55.

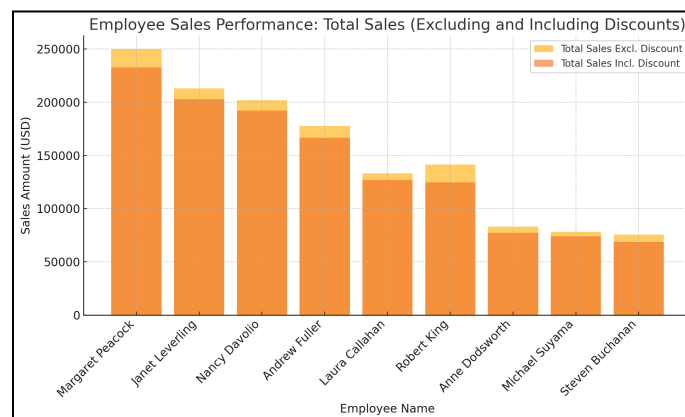
Table 9. Performance of Employees based on Total Sales Amount, including Discounts

full_name	job_title	total_sales_excl_discount	total_unique_orders	total_orders	avg_product_amount_excl_discount	avg_order_amount_excl_discount	total_discount_amount	total_sales_incl_discount	total_discount_percentage
Margaret Peacock	Sales Representative	250187.45	156	420	595.68	1603.77	17296.6	232890.85	6.91
Janet Leverling	Sales Representative	213051.3	127	321	663.71	1677.57	10238.46	202812.84	4.81
Nancy Davolio	Sales Representative	202143.71	123	345	585.92	1643.44	10036.11	192107.6	4.96
Andrew Fuller	Vice President, Sales	177749.26	96	241	737.55	1851.55	11211.51	166537.76	6.31
Laura Callahan	Inside Sales Coordinator	133301.03	104	260	512.7	1281.74	6438.75	126862.28	4.83
Robert King	Sales Representative	141295.99	72	176	802.82	1962.44	16727.76	124568.23	11.84
Anne Dodsworth	Sales Representative	82964	43	107	775.36	1929.4	5655.93	77308.07	6.82
Michael Suyama	Sales Representative	78198.1	67	168	465.46	1167.14	4284.97	73913.13	5.48
Steven Buchanan	Sales Manager	75567.75	42	117	645.88	1799.23	6775.47	68792.28	8.97

c. Key insights and Findings

- **Top Performers:** Based on total sales, both before and after discounts, Margaret Peacock, Janet Leverling, and Nancy Davolio are the highest performing sales representatives. They manage a huge volume of orders, which greatly increases the business's income.
- **High Average Order Value:** Employees who handle more complex or higher-value sales, such as Andrew Fuller, typically have higher average order values than those who handle fewer orders.
- **Impact of Discount:** Each employee's overall discount percentage is different. Compared to other sellers, Robert King gives a comparatively higher discount percentage (11.84%), than Margaret Peacock, the top seller, which could be a calculated tactic to complete purchases.
- **Balanced Performance:** Employees with more balanced data, such as Laura Callahan's moderate sales, average order value, and discount percentage, show a well-rounded performance.

Figure 10. Employee Sales Performance: Total Sales (Excluding and including discounts)



The graph in Figure 10 makes it simple to see which workers are producing the most sales and how much of an impact discounts have on their overall sales.



13. Business Question 10

a. Description

The sales team aims to evaluate each employee's contribution to the overall sales of each product category in order to assess their success in those categories. The full name of the employee, the total sales amount (discounts included), the percentage of each employee's sales within that category relative to their overall sales, and the percentage of each employee's sales within the category relative to the total sales of the category are among the requested KPIs. The team can quickly determine the top contributors in each category by sorting the data according to category name and total sales. The following metrics are needed for every employee:

- Category Name: The category for which the employee's sales are being measured.
- Full Name: The employee's name (first and last combined).
- Total Sales (Including Discount): The total sales amount for the employee within the category.
- Percentage of Employee's Sales within Category: The percentage of the employee's total sales that come from this category.
- Percentage of Total Sales within Category: The employee's contribution to the total sales of the category.

b. Results

The analysis in Table 10 allows the sales team the following insights:

- With sales of \$52,842.35, Margaret Peacock is the category leader in beverages. Her sales account for 21% of her entire sales as well as more than 18% of the category's total sales.
- Nancy Davolio is the second-highest salesperson in this area, with beverages accounting for 24.15% of her total sales.

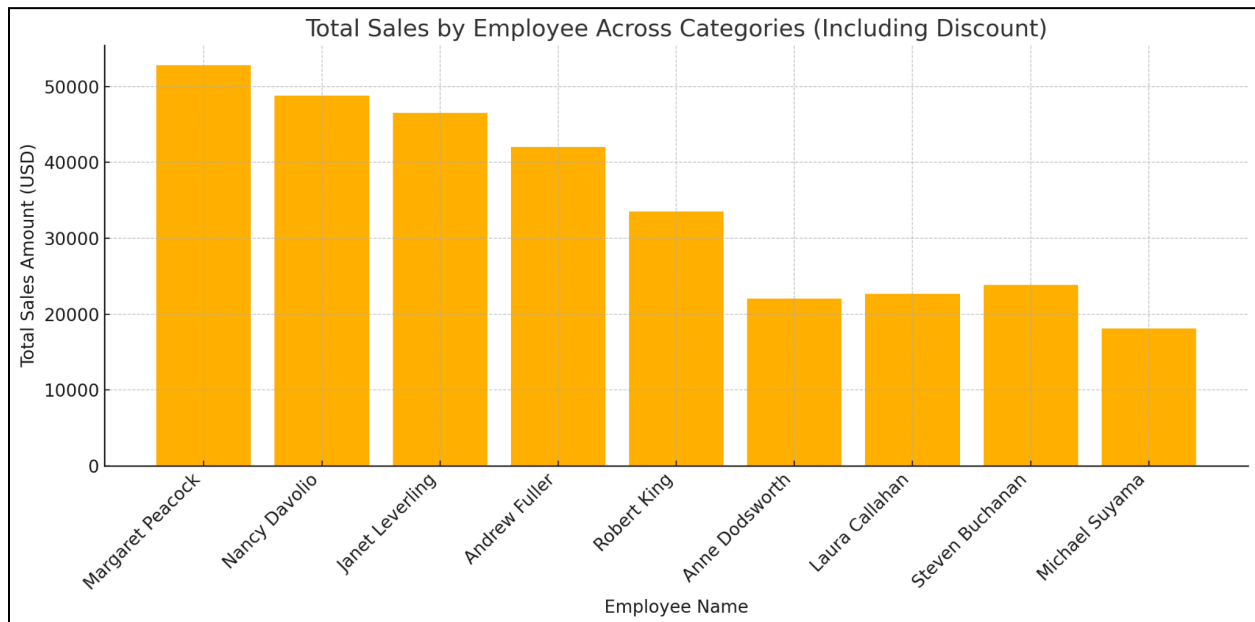
Table 10. Employee performance in categories with respective total sales within each category

category_name	full_name	total_sales_incl_discount	pct_of_employee_sales	pct_of_category_sales
Beverages	Margaret Peacock	52842.35	21.1211	18.44237
Beverages	Nancy Davolio	48832	24.15707	17.04272
Beverages	Janet Leverling	46506.55	21.8288	16.23113
Beverages	Andrew Fuller	42029.4	23.64533	14.66857
Beverages	Robert King	33517	23.72113	11.69768
Beverages	Anne Dodsworth	20513.6	24.72591	7.1594
Beverages	Laura Callahan	18640.8	13.98399	6.50578
Beverages	Steven Buchanan	13517.5	17.88792	4.71771
Beverages	Michael Suyama	10127.75	12.9514	3.53466
Condiments	Margaret Peacock	25242.45	10.08941	22.20195
Condiments	Andrew Fuller	16353.1	9.20009	14.38334
Condiments	Laura Callahan	15447.95	11.58877	13.58721
Condiments	Janet Leverling	14109.15	6.62242	12.40968
Condiments	Nancy Davolio	14033.85	6.94251	12.34345
Condiments	Anne Dodsworth	11106.9	13.38761	9.76905
Condiments	Robert King	9668.7	6.84287	8.50409
Condiments	Michael Suyama	4930.2	6.30476	4.33635
Condiments	Steven Buchanan	2802.45	3.70853	2.46489
Confections	Janet Leverling	34859.31	16.36193	19.6835
Confections	Nancy Davolio	30343.54	15.01088	17.13365
Confections	Margaret Peacock	29543.06	11.80837	16.68165
Confections	Laura Callahan	22639.5	16.98374	12.78352
Confections	Andrew Fuller	21696.95	12.20649	12.2513
Confections	Robert King	16754.45	11.8577	9.46049
Confections	Anne Dodsworth	8782.94	10.58645	4.95934
Confections	Michael Suyama	6990.3	8.93922	3.94711
Confections	Steven Buchanan	5489.05	7.26375	3.09942
Dairy Products	Nancy Davolio	38012.8	18.80484	15.12463
Dairy Products	Margaret Peacock	37047.6	14.80794	14.74059
Dairy Products	Janet Leverling	34546.3	16.21501	13.74537
Dairy Products	Robert King	30581.5	21.64357	12.16784
Dairy Products	Andrew Fuller	25594.7	14.39933	10.18368
Dairy Products	Steven Buchanan	23850.4	31.56161	9.48966
Dairy Products	Anne Dodsworth	22017.9	26.5391	8.76054
Dairy Products	Laura Callahan	21596.2	16.20108	8.59275
Dairy Products	Michael Suyama	18083.1	23.12473	7.19495
Grains/Cereals	Margaret Peacock	23704.35	9.47464	23.53331
Grains/Cereals	Janet Leverling	21898.15	10.27835	21.74014
Grains/Cereals	Andrew Fuller	12063.1	6.78658	11.97606
Grains/Cereals	Laura Callahan	11723.9	8.79506	11.63931
Grains/Cereals	Michael Suyama	9951.1	12.7255	9.8793
Grains/Cereals	Nancy Davolio	9173.95	4.53833	9.10775
Grains/Cereals	Robert King	6673.5	4.72306	6.62535
Grains/Cereals	Steven Buchanan	4233.25	5.60193	4.2027
Grains/Cereals	Anne Dodsworth	1305.5	1.57357	1.29608
Meat/Poultry	Margaret Peacock	34179.22	13.66144	19.18146
Meat/Poultry	Andrew Fuller	32973.51	18.55058	18.50482
Meat/Poultry	Robert King	24939.24	17.65035	13.99596
Meat/Poultry	Janet Leverling	20921.7	9.82003	11.74131
Meat/Poultry	Laura Callahan	17657.5	13.24633	9.90943
Meat/Poultry	Nancy Davolio	16346.07	8.08636	9.17346
Meat/Poultry	Steven Buchanan	11869.4	15.70696	6.66114
Meat/Poultry	Anne Dodsworth	9920.36	11.95743	5.56733
Meat/Poultry	Michael Suyama	9381.8	11.99748	5.26509
Produce	Nancy Davolio	20454.45	10.11877	19.43072
Produce	Margaret Peacock	18586.75	7.42913	17.6565
Produce	Janet Leverling	12718.15	5.96952	12.08162
Produce	Michael Suyama	12323.9	15.75985	11.7071
Produce	Laura Callahan	12146.8	9.11231	11.53886
Produce	Robert King	11552.4	8.17603	10.97421
Produce	Andrew Fuller	9476.8	5.33156	9.00249
Produce	Steven Buchanan	7605.1	10.06395	7.22447
Produce	Anne Dodsworth	404.25	0.48726	0.38402
Seafood	Margaret Peacock	29041.67	11.60796	20.50631
Seafood	Janet Leverling	27491.99	12.90393	19.41208
Seafood	Nancy Davolio	24947.05	12.34124	17.6151
Seafood	Andrew Fuller	17561.7	9.88004	12.40031
Seafood	Laura Callahan	13448.38	10.08873	9.4959
Seafood	Anne Dodsworth	8912.55	10.74267	6.29315
Seafood	Robert King	7609.2	5.38529	5.37285
Seafood	Michael Suyama	6409.95	8.19707	4.52606
Seafood	Steven Buchanan	6200.6	8.20535	4.37824

c. Key insights and Findings

- **Top Performers by Category:** In the 'Beverages' category, employees like Margaret Peacock and Nancy Davolio are among the top contributors, and Andrew Fuller and Janet Leverling also make noteworthy contributions.
- **Category Specialization:** An employee's specialization is indicated by the proportion of sales they generate inside a particular category. For example, Nancy Davolio's sales in the 'Beverages' category account for 24.15% of her overall sales, indicating a high emphasis on that product line.
- **Category Contribution:** Margaret Peacock plays a significant role in driving revenue in the 'Beverages' category, accounting for more than 18% of total sales.

Figure 11. Total Sales by Employee Across Categories (Including Discount)



The total sales (discounts included) generated by each employee across several categories are shown in the graph above in Figure 11. The sales team can rapidly compare employee performance and their percentage of total sales within each category with this picture.



14. Conclusion

The project effectively fulfilled the needs of the stakeholders by offering a thorough study of the company's sales, pricing, and operational performance across many business units. We gave the price, logistics, and sales teams useful information through a series of data-driven inquiries, empowering them to decide on product price, staff performance, and sales tactics that are tailored to particular categories.

- **Key Findings, Insights, and Outcomes:**

- Pricing Performance: The study showed that there is a significant amount of price variation among products in various categories. Because outliers like Côte de Blaye are priced far higher than usual for their category, it is important to investigate whether their premium status is in line with consumer demand. The analysis also showed how many products have prices below the median, indicating that the business would profit from reevaluating its consumer targeting and product positioning plans.
- Logistics and Supplier Management: Product availability, freight expenses, and supplier area performance were all clearly visible to the logistics team. They can improve supply chain operations to guarantee smoother inventory movement by knowing how stock levels are distributed throughout areas, particularly for high-demand categories like beverages and condiments.
- KPIs and Employee Performance: Based on overall sales and order quantities, the sales team can be able to identify their top achievers, including Janet Leverling, Andrew Fuller, and Margaret Peacock. A deeper comprehension of their sales methods can be gained by gaining insights into the discount strategies employed by employees such as Robert King, who provides somewhat larger discounts. Furthermore, the category-specific sales performance research revealed that some employees—Nancy Davolio, for example—have crucial product lines like beverages as their areas of expertise, which greatly influences their performance.

- **Success in achieving goals:**

Stakeholders' company concerns have been effectively addressed by the project, yielding insightful answers that support the strategic goals of the organization. All three departments—Sales, Logistics, and Pricing—had access to the information they needed to



make better decisions. For instance, the sales team can use performance KPIs to enhance employee incentives and training, while the pricing team can now assess product price more efficiently. On the basis of current information regarding supplier performance and freight expenses, the logistics team may more effectively manage their regional stock levels.

- **Future Work and Recommendations:**

- Pricing Adjustments: It is advised that the Pricing Team keep a close eye on any products that are priced significantly above or below the category average, as these are known as pricing outliers, in each category. Furthermore, additional customer segment analysis might point to areas where items should be repositioned to better meet consumer expectations.
- Sales Incentives: The sales team should think about implementing focused reward programs based on the KPIs from this study. Acknowledging exceptional workers like Margaret Peacock and offering training to staff members with lower average sales per order could maximize the performance of the team as a whole.
- Supply Chain Optimization: To guarantee the best possible distribution, the logistics team should keep a close eye on the stock levels in each region. Adding supplier lead times and shipment efficiency to the research could aid in locating possible supply chain improvement areas.

In conclusion, this initiative has given stakeholders a better understanding of their company's performance, answered important queries, and offered guidance for future expansion. The organization can improve its pricing tactics, increase sales performance, and streamline operations by putting the analysis's recommendations into practice.



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